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PALEOMAGNETISM AND CROSS-FOLDING IN A KEY AREA OF THE ASTURIAN ARC (SPAIN)

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Abstract. In the northwest corner of Spain, Paleozoic and older rocks outcrop in an arcuate pattern (Asturian Arc), concave toward the east. The arc may be primary (before sedimentation) or secondary (after sedimentation). Supporting a secondary origin are many structural features, including two sets of folds, one radial and one tangential to the arc. The Villasecino area was selected for a pilot study of the relationships between paleomagnetism and folding. Detailed structural mapping has shown that radial folds postdate tangential folds. Interference of the two sets has produced the doubly plunging Villasecino anticline. Forty two oriented paleomagnetic cores were drilled at six sites distributed around the anticline, all within red limestones (griotte) of the Alba formation (lower Carboniferous). The typical sample contains 95% by volume of fine-grained calcite; each grain being rimmed by very fine-grained goethite and other iron-bearing minerals. The natural remanent magnetism (NRM) is of moderate intensity (about 3×10^{-3} A m^{-1}) readily measured and very homogeneous for each site. Within-site scatter of vectors is low, but the site mean varies significantly from site to site. The angle between bedding and magnetization vector is nearly constant. These data suggest that magnetization predated folding. Thirteen samples from six sites were subjected to alternating field demagnetization up to 50 mT, and this showed the NRM to be very stable. Thermal demagnetization indicated a unique stable component with blocking temperature in the range 300°-600°C. Chemical demagnetization was attempted using sodium sulphite. Results suggest that the carriers of magnetization are not iron hydroxides but probably hematite. Even after demagnetization, a simple bedding correction does not yield a good grouping of magnetization vectors. We have therefore tested five other, more complex models of unfolding, based on structural observations. The statistically best grouping is produced by a model involving two successive fold phases, where first folds have horizontal E-W axes and second folds have vertical axes. This is compatible with structural evidence. After unfolding, we obtain D = 101.8°, I = 13.1° (α_{95}° = 1.5°) for the orientation of the earth's magnetic field in the area. If we compare with the direction for the Carboniferous of 'stable Europe' we get a difference in declination of 50° to 65° according to the model used for closing the Bay of Biscay, possibly in association with tectonic evolution of the Asturian Arc.

Introduction : The Asturian Arc

In the northwestern corner of Spain (Galicia and Asturias), Paleozoic and older rocks outcrop in an arcuate pattern, concave toward the east (Figure 1). This Asturian Arc was first shown on maps by Schulz [1858] and Suess [1885]. Ever since, there have been many attempts to explain its origin. Those who favor a primary (presedimentation) origin have been outnumbered by proponents of a secondary (tectonic) origin.

Supporting a primary origin are studies of sedimentary environments and paleogeography in the Cantabrian Zone (Figure 1b) where the arc is tightest [Zamarreño, 1972]. Nevertheless, a number of structural features lie parallel to the arc: these include traces of major thrusts, fault traces, fold hinges, traces of axial surfaces of folds, and, in some lithologies, tectonic cleavage. Such features support a tectonic or secondary origin for the arc [de Sitter, 1961; Matte, 1968; Julivert, 1971b]. It is, of course, possible that the arc is indeed partly primary but that secondary tectonic displacements have increased the curvature. Thus Ries and Shackleton [1976] have proposed that the arc is an orocline [Carey, 1955] formed by Variscan bending about a vertical axis of an originally straight or much less curved fold belt. In contrast, Matte and Ribeiro [1975] have suggested that secondary curvature was produced by relative displacement toward the west of a Cantabrian microplate. Also, deformation in north Spain may not be independent of Hercynian deformation in Brittany. Since the time of Choubert [1935] many hypotheses have been made concerning the existence of an Iberian-Armorican Arc [see, for example, Matte and Ribeiro, 1975].

The various models of the Asturian Arc can, in principle, be checked using methods sensitive to any of the components of permanent deformation, these being rigid translations, rigid rotations, and strains. Permanent strains of large magnitude are recorded in rocks by deformed fossils and other objects as well as by structures such as folds. Some progress has been made in measuring strains throughout the Asturian Arc [Matte, 1968; Ries and Shackleton, 1976], but major obstacles are the absence of good strain markers and the strongly heterogeneous nature of the strain.

For determining components of rigid motions, by far the best tool would appear to be paelomagnetism, especially if applied to rheologically resistant lithologies which have undergone little internal strain. Under these conditions it is possible to determine (1) rigid rotations about

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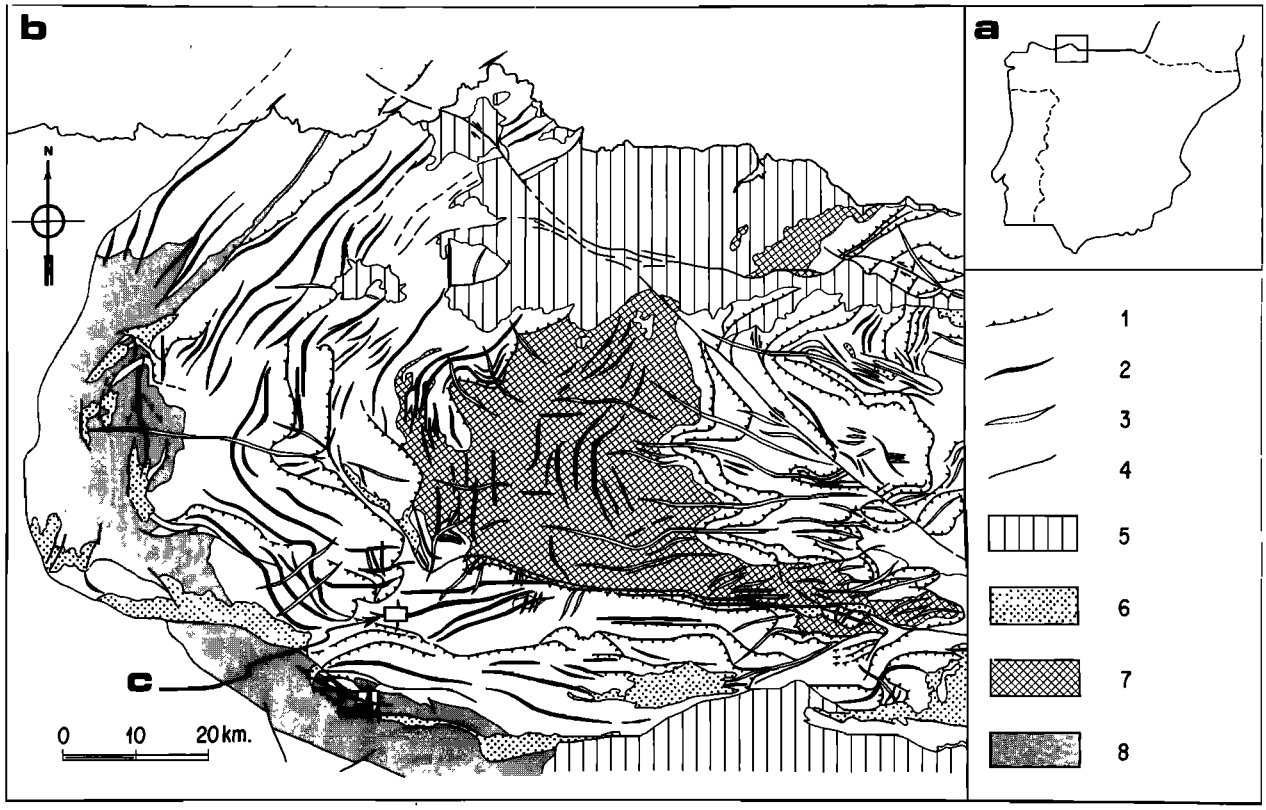


Fig. 1. (a) The Asturian Arc of NW Spain and its structural elements (modified [after Julivert and Marcos, 1973]). (b) In the Cantabrian zone, Precambrian rocks of the Narcea Antiform (8) are overlain by a Paleozoic sequence, culminating in Carboniferous shales of the Central Coal Basin (7) and unconformable Stephanian sediments (6). Below the Stephanian, the area is affected by thrusts (1), faults (4) and two generations of folds, one with axial surfaces (2) tangential to the arc, the other with radial axial surfaces (3).

vertical and horizontal axes and (2) large changes in geographical latitude. A test for the former is of great importance in discriminating between models for a primary or secondary origin of the Asturian Arc.

Déformation in the Cantabrian Zone

In the Cantabrian Zone (Figure 1b), Paleozoic sediments are of low metamorphic grade. Penetrative internal strain is restricted to a few lithologies (shales), and most of the regional deformation has occurred by development of faults, overthrusts, nappes, and folds. The structural history of the region has been summarized by Julivert [1971b].

Overthrusts are common throughout the Cantabrian Zone [Julivert, 1971a]. Overthrust sheets have complex shapes and are stacked multiply, producing many repetitions in the stratigraphic sequence. Most of the thrust surfaces are also folded, and it is likely that much of the thrusting preceded the main episodes of folding. There is evidence that thrust sheets were translated radially inward toward the center of the arc. It seems likely that relative rotations have occurred between individual thrust sheets, as reported elsewhere by Grubbs and Van der Voo [1976].

Folding occurred under conditions favoring flexural slip [de Sitter, 1961; Julivert and Marcos, 1973]. This means that rheologically resistant beds have been buckled without large changes in arc length and with a minimum of internal strain. Such layers are well suited to paleomagnetic sampling. Contact between neighboring resistant layers has been maintained, either by relative slip on bedding discontinuities or by shear distributed throughout less resistant layers (for example, shales).

Most folds have horizontal or nearly horizontal hinges and vertical or almost vertical axial surfaces. They may be grouped into two large families (Figure 1b), one with hinges radial, the other with hinges tangential, to the Asturian Arc [Julivert and Marcos, 1973]. Where radial and tangential folds intersect, they produce interference patterns. Some of these are of dome-and-basin type, from which it is difficult to draw conclusions concerning the relative ages of the radial and tangential components. In other examples, however, it is clear that axial surfaces of tangential folds are refolded about vertical axes lying in the axial surfaces of radial folds. From such relationships, Julivert and Marcos have deduced that most of the radial folds formed later than the tangential ones. Our own structural investigations wholly support this interpretation. We have found instances where a tectonic cleavage associated with tangential folds is itself refolded around radial folds.

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In carbonate-rich rocks, flexural slip has led to formation of imbricate calcite fibers in gaps between bedding surfaces. These fibers give the sense as well as the direction of slip [Durney and Ramsay, 1973]. Associated with interference structures in Cantabria are two well defined and separate sets of fibers, one approximately normal to radial fold axes, the other nearly normal to tangential axes. We take this as evidence that the two fold sets formed independently in time. Furthermore, on any bedding-surface gap, those fibers found closest to the walls of the gap are, in most if not all examples, normal to tangential fold axes; thus we conclude that tangential folds (henceforth labeled F1) were succeeded in time by radial folds (F2) in accordance with the interpretation of Julivert and Marcos [1973].

For our pilot study of paleomagnetism and folding, we chose a small area (labeled c in Figure 1b and shown enlarged in Figure 2) containing an interference structure (the doubly plunging Villasecino anticline) and lower Carboniferous limestones (Alba formation) which were relatively resistant to deformation and contain magnetic minerals. The area was also selected because it is here that the arc begins its swing northward from an east-west attitude (Figure 1b).

Cross-Folding in the Villasecino Area

The Villasecino area (Figure 2) is here taken to be a small area (about 3.5 x 1 km) centered around the village of Villasecino (6°00'W, 42°57'N). The general geology of this area and surrounding upper reaches of the Luna and Sil valleys have been described by Van den Bosh [1969] in a comprehensive paper accompanied by a geological map at 1/50,000. Prominent on this map and on our own (Figure 2a) is the Villasecino anticline, a noncylindrical structure with a hinge essentially horizontal, east-west and parallel to other F1 folds. In the Villasecino area, it affects Devonian and Carboniferous sediments, including red limestones (griottes) of the Alba formation. Local topographic relief is moderate, so that traces of bedding at outcrop (Figure 2b), detailed measurements of dip and strike (Figure 2c), and stereographic analysis of these structural data (Figure 2d) can be used to reconstruct a horizontal section through the area (Figure 2e).

Detailed stereographic analysis (Figure 2d) illustrates several important points. Thus a plot of all the bedding normals (all areas, Figure 2d) shows how noncylindrical the major fold is. Areas of maximum point density represent the northern and southern limbs plus the eastern and western hinge zones. The total area has been divided into four subareas for further analysis.

1. Subarea A (Figure 2d) is more cylindrical than the total structure: the average fold axis plunges at about 20° to the WNW and the axial surface (a) is steep. The northern (n) and southern (s) limbs nevertheless show a scatter of points which we interpret as due to F2 refolding. Further evidence for this comes from a study of calcite fibers freshly exposed on bedding surfaces in a road-cut across the hinge of the major fold (section S, Figure 2b). From the orientation of these fibers and the sense of slip, determined in the field, one can deduce the axis and sense of the associated rotation of the bedding during folding. This rotation is, of course, only incremental, as the fibers do not record the entire history of folding. The instantaneous rotation paths of bedding normals (Figure 2d, area A, tails to open circles) show how refolding has occured on the limbs of the major anticline. The tails point away from the great circle (p) that would represent a perfectly cylindrical fold. Some tails are almost normal to p, indicating refolding about minor F2 axes that lie within p.

2. Subarea B (Figure 2c) is larger in extent. The average F1 axial surface trends more nearly east-west. The amount of refolding on the limbs is greater, as witnessed by the scatter of bedding normals (Figure 2d).

3. In subarea C the F1 axis is approximately horizontal. Refolding is strong and affects the hinge as well as the limbs of the major anticline.

4. Subarea D (Figures 2c and 2d) is a small-scale dome, with axes plunging approximately north, south, east, and west.

In conclusion, stereographic analysis supports earlier contentions about the order of fold events. It shows how the major anticline (F1) has been modified by later folds (F2), undergoing in the process changes in (1) the angle of plunge of the hinge, (2) the direction of plunge of the hinge, (3) the trend of the axial surface, and (4) the planarity of the limbs. These changes vary from subarea to subarea, F2 folds existing on a scale smaller than the F1 antiform. Both types of interference patterns described by Julivert and Marcos [1973] occur here in one small area. Thus the area is potentially useful for studying the effects of cross folding on paleomagnetism.

Paleomagnetic Samples

The Alba Formation

In the Villasecino area the Alba formation (of Viséan age) attains a maximum thickness of about 13 m. It consists of red nodular limestones (griottes) pink to grey nodular limestones, red shales, red siltstones, and bedded red chert with radiolarians [Van den Bosch, 1969]. The original sediments were probably deposited on a flat lying topography, in a shallow marine environment, under very quiet oxidizing conditions. Conditions of diagenesis are less well known.

Sampling Procedure

To avoid the effects of strain as much as possible, sampling was restricted to the main band of red griottes. A total of 42 oriented paleomagnetic cores were drilled at six sites distributed over the Villasecino area (Figure 2b). Three of the sites (2A, 19, 21) are on the northern limb of the main anticline, the other three (2B, 20, 22) on the southern limb. Sites were chosen on the basis of structural position, outcrop, and accessibility. Thus sites 20 and 21 in the west lie within subarea A, where the F1 axial trace is WNW; site 19 lies in subarea C, where the F1 trace is nearly EW; whereas sites 2A, 2B, and 22 lie in subarea D, where the major anticlinal hinge is bent upward into a local dome.

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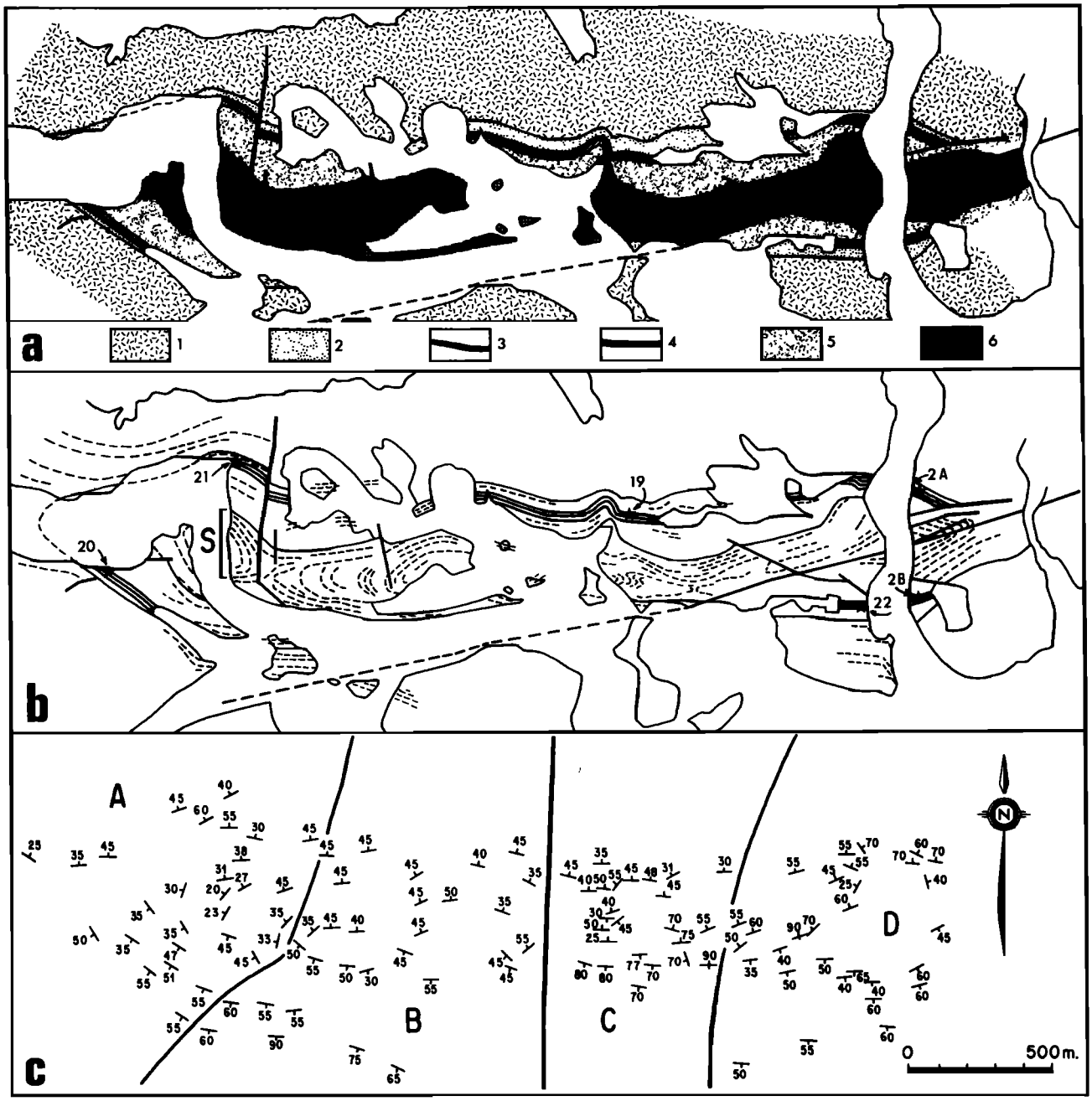


Fig. 2. Structure of the Villasecino area. (a) outcrop map showing Devonian limestones of Candas formation (6); lower Carboniferous rocks including, sandstones of Ermita formation (5); limestones of upper part of Ermita formation (4); limestone of Alba formation (3); platy limestones (2) and thicklybedded limestones (1) of Caliza de Montana Formation. Thickest lines are fault traces. (b) Bedding traces (dashed lines) and numbered paleomagnetic localities (stars). (c) Strike and dip of bedding, with subareas A, B, C, D. (d) Stereographic projection (lower hemisphere, equal area) of bedding normals (dots and open circles) minor fold axes (triangles) and major fold axes (squares). (e) Reconstructed horizontal section through area, showing trace of bedding (solid curves) and its dip (in degrees), axial traces of tangential folds (1) and of radial folds (2).

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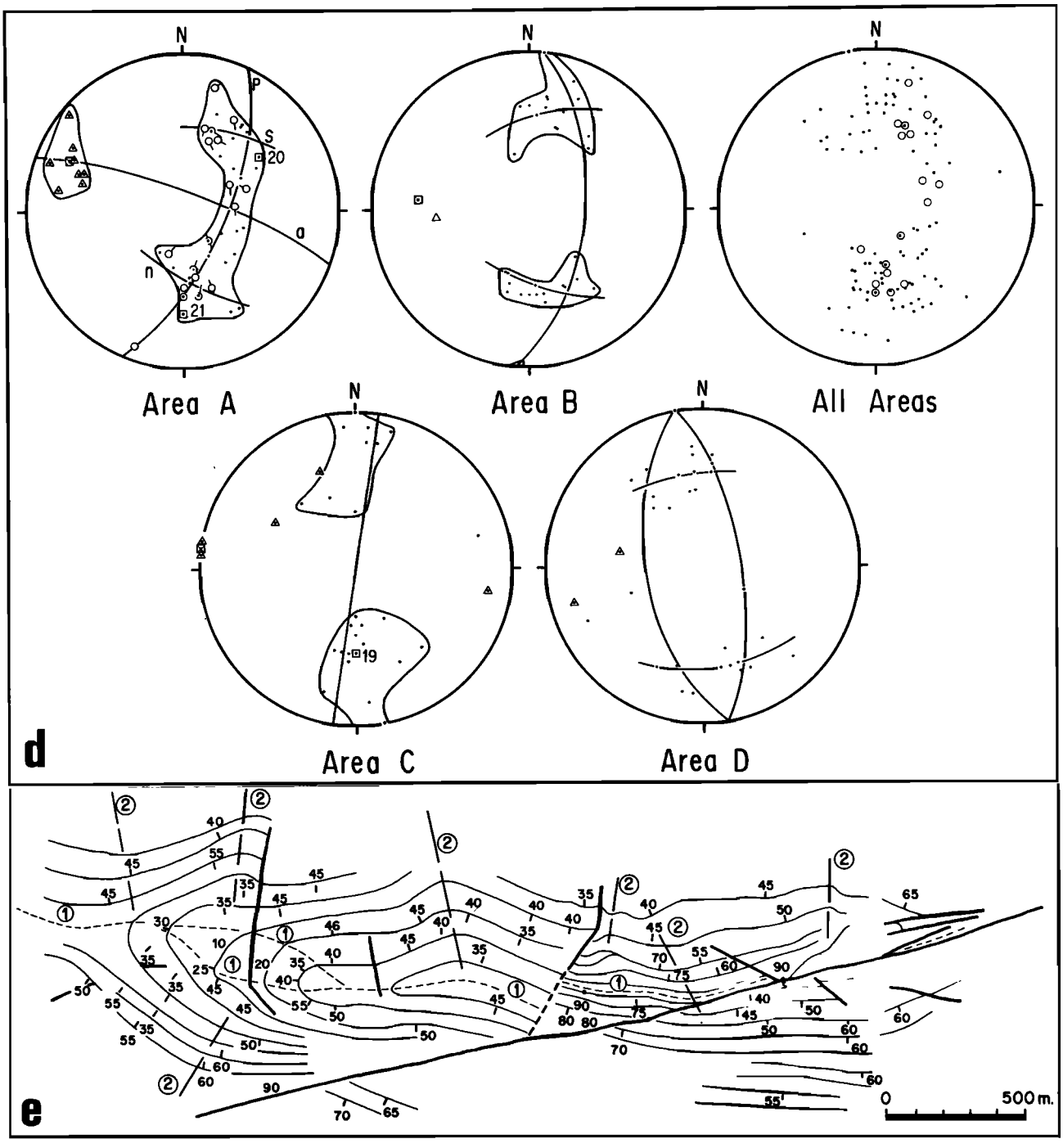


Fig. 2. (continued)

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At each site, six to eight oriented cores were drilled within a radius of about 5 m. The orientation of bedding was carefully measured for each core; where such a measurement was not possible, no core was drilled. Sampling was restricted to the most massive rock with a minimum of tectonic or pretectonic cleavage and stylolites. From this field collection, only one specimen per core was available for study.

Sample Composition

The mineralogical and chemical composition of samples was studied using an optical microscope (transmitted and reflected light), X ray analysis and microprobe analysis.

1. Under the optical microscope (transmitted light), about 95% by volume of the rock consists of equant calcite grains (grain size about 2 μm). These are coated with even finer-grained films of opaque minerals (about 5% by volume). Scattered within the matrix are fossil fragments of various sizes (5% by volume, mainly calcite).
2. In reflected light, polished sections reveal that opaque minerals are predominantly goethite. These rim the calcite grains and also occur in small veins and spots throughout the matrix.
3. X ray analysis reveals the presence of calcite, with minor amounts of goethite.
4. Microprobe analyses for calcium and iron show that the calcite grains are iron-free. The iron is concentrated in very thin layers (<1 μm), calcium-free, where the grain size is very much less than 1 μm.

Internal Deformation

Little is known about the effects of strain upon the orientation of NRM in real rocks, but it is reasonable to assume that magnetization vectors might rotate by as much as any passive marker would [Kodama and Cox, 1978]. Depending upon the orientation of the strains, the magnetization could be significantly modified not only in inclination but also in declination. For areas such as Cantabria it is therefore important to be able to estimate the strains and their possible effects on magnetization.

Experiments performed on artificial materials have yielded results which at first sight may appear contradictory but in detail are not. Thus Kodama and Cox [1978] reported little change in magnetization direction of clay samples after bulk shortening of up to 33% but the mechanism of deformation was shown to be shearing alongs narrow conjugate zones involving only a few percent by volume of each sample. In contrast, Ozima [1980] obtained rotations of up to 20° for equivalent shortening, but here deformation was intracrystalline and caused by rotation of ellipsoidal magnetic grains throughout the sample. We may therefore conclude that whether or not the strain effect is important depends upon the mechanism of deformation and the scale at which it occurs, relative to the nature and size of magnetic minerals.

Bearing these points in mind, we have examined thin sections of Alba griotte under the optical microscope, with the following results:

1. At the grain scale, fossil fragments appear undistorted with uncorroded boundaries. Most fragments are single crystals of calcite and show little or no twinning. In the matrix, calcite grains are equant and apparently of sedimentary origin; thus they do not appear to have been deformed. We conclude that intracrystalline deformation is negligeable.
2. At the sample scale there is evidence for deformation. Some samples show extension veins infilled with crystalline calcite. In two samples only, these veins are very slightly buckled. More common are seams of opaque material, up to 40 μm thick and spaced at intervals of more than 1 cm. These seams truncate fossil fragments and have stylolitic forms. They may be identified as zones of localized shortening with volume loss due to removal of carbonate material. Such zones are common in limestone (especially griottes) and are attributed to stress-induced diffusion. In the Alba griottes, most seams are parallel to bedding and probably formed during burial, but others are equivalent to tectonic cleavage and are probably contemporaneous with folding. From observation of displaced fossil margins we have estimated the shortening across individual seams as being no more than 1 mm, usually much less. Thus the resulting bulk shortening of the rock is about 10% or less. Also the seams at present account for less than one hundredth of the total rock volume. Since, however, magnetic minerals are preferentially concentrated along the seams, the latter may perhaps contribute significantly to the bulk NRM. We have not been able to investigate this possibility directly because the seams are too closely spaced compared with the size of paleomagnetic samples.

Paleomagnetism

Natural Remanent Magnetism (NRM)

For 42 standard specimens the direction and intensity of NRM was measured using a computerized (PDP 11-05) flux gate spinner magnetometer (Schonstedt DSM-1). The NRM is of moderate intensity and readily measured (Figure 3a, inset). The main intensity is 3.3×10^{-3} A m^{-1} (3.3×10^{-6} emu cm^{-3}) with standard deviation of 0.96×10^{-3} A m^{-1}. Site mean directions of NRM have been calculated with α_{95} circles of confidence (Figure 3a and Table 2, row 1). The data can be summarized as follows:

1. Within-site scatter is quite low. This probably reflects the 5 m radius of sampling, within which bedding orientation does not vary greatly.
2. The site mean orientation varies significantly from site to site.
3. Site mean orientations are all different from that of the earth's magnetic field at the present time. This suggests a stable type of magnetization.

A preliminary comparison with structural data suggests a prefolding origin for magnetization. For example, site means 2B and 22 are well-grouped and both lie on the southern limb of the major F1 anticline; similarly, 2A and 19 are well grouped and lie on the northern limb. Also site means 2A, 19, 21, which all belong to the same northern limb, show differences in declination of NRM which will be correlated later with the

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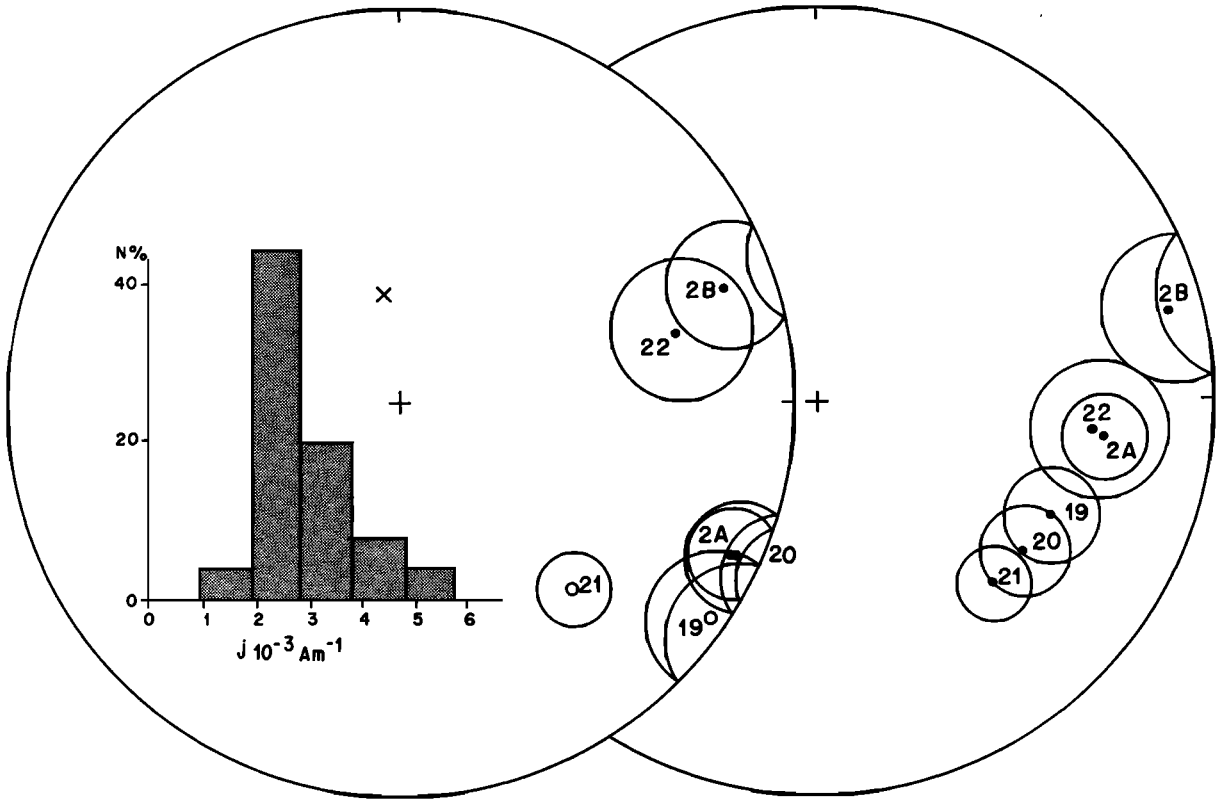


Fig. 3. Stereographic projection of NRM vectors before (left hand side) and after (right hand side) correction for dip of bedding. Site means (dots = lower hemisphere, open circles = upper hemisphere) with corresponding site numbers are surrounded by their α_{95} circles of confidence. Inset (left) is frequency plot for magnetization intensity.

changing orientation of the F1 fold axis (Table 1).

A very simple but important test can be done to check whether magnetization is acquired before folding. If beds undergo rigid rotations with no internal strain, the angle between the NRM vector and the bedding surface (that is, the original inclination) must remain constant. For the 42 standard specimens from the Villasecino area the angle in question has a mean of 18.3°, with a standard deviation of 13.5°. The mean value is very close to the inclination of the earth's field during the lower Carboniferous (as discussed later). The small deviation suggests that there has been little or no later remagnetization (since internal strains are known to be small). This has been checked using demagnetization techniques (described in the following sections).

After NRM measurements, because of a lack of specimens, samples were distributed into three groups for demagnetization experiments and microscope work. Thirteen samples were progressively demagnetized using alternating fields, 17 were thermally demagnetized, and 10 were used in attempts at chemical demagnetization.

Alternating Field (af) Demagnetization

The NRM of 13 samples (about two per site) from all six sites was measured after each of five steps of 10 mT applied in a demagnetizing system (Schonstedt GSD-1). To check the higher part of the coercivity spectrum, six samples were demagnetized up to 100 mT.

Curves of NRM intensity versus applied field (Figure 4a) have similar forms for all samples and sites. At 50 mT the intensity is still 60% to 80% of the initial value. The medium destructive field (MDF, corresponding to a 50% decrease of initial NRM) has a minimum value of 70 mT; for most samples, values of the order of 100 mT can be estimated from the trends of the curves. Zijderveld diagrams [Zijderveld, 1967] for two typical samples (Figure 5a) show that the NRM directions do not change greatly during cleaning up to 40 mT. There is a general decrease toward the origin after removal of a low coercivity component (0-20 mT). The characteristic direction for each curve is obtained using vectorial subtraction in the range 30-40 mT; above 40 mT, small spurious components generally appear, giving an increase in the scatter of directions for each site. These components are interpreted as secondary ones acquired during af treatment, as they behave erratically.

Because few samples were demagnetized for each site, it was not possible to check the efficiency of af cleaning using Fisher statistics on each site before and after cleaning (see Table 3). One may, of course, use the 13-sample characteristic directions as a single population, but here any slight improvement coming from cleaning will be totally hidden due to complex folding of the Villasecino structure (Figure 3a). The effi-

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TABLE 1. Structural Data Averaged for Each of the Six Sites Around the Villasecino Anticline

Sites	Total Number of Samples Used*	Bedding		F1 Fold Axis		F2 Fold Axis	
		Dip	Strike	Plunge	Strike	Plunge	Strike
2a	6 (1, 3)	70°	277°	0°	90°	46°	0°
2b	7 (3, 3)	54°	76°	0°	90°	40°	180°
19	7 (1, 3)	45°	276°	0°	90°	40°	0°
20	7 (2, 3)	50°	147°	22°	305°	-	-
21	7 (3, 2)	60°	272°	22°	305°	-	-
22	8 (3, 2)	60°	87°	0°	90°	40°	180°

*Two successive figures in brackets indicate number of samples used in af and thermal cleaning, respectively.

ciency of cleaning will therefore be checked on the whole population of cleaned samples (either af or thermal) but after correction for folding. This will be done using the unfolding model F described later.

Thermal Demagnetization

To check the stability of the NRM further, a majority of samples (17) belonging to the six sites were chosen for thermal demagnetization using a Schonstedt TSD-1 system, where samples are heated and cooled in a zero field (nonmagnetic furnace with mu-metal magnetic shielding). Measurements of NRM were made after each of eight steps between 20° and 630°C.

The decrease in intensity with temperature (Figure 4c) is similar for all 17 samples and marked by (1) an apparent increase of intensity up to 150°C, (2) a rapid decrease of intensity in the range 450°-510°C, indicating a main range of blocking temperatures, and (3) no decrease or an evenly increasing intensity in the range 510°-630°C.

These results can be closely related to variations of NRM directions. Two characteristic samples (Zijderveld diagrams, Figure 5b) illustrate the following results:

1. At 150°C there is removal of a northerly component with positive inclination for 'in situ' (geographic) coordinates. This component is close to the present earth's field direction at the site and is interpreted as being recent viscous magnetization.

2. After removal of this component the direction stabilizes up to 550°C. The characteristic direction for this stable magnetization has been obtained from the 300°C data, which give the lowest within-site scatter.

3. For a very few samples the direction can be followed up to 630°C (Figure 5b), but for most samples spurious components may appear from around 450°C onward, mainly in the range 510° to 630°, where the intensity may even increase slightly.

4. There is no evidence for a higher temperature component with coherent direction in the formation either before or after unfolding.

From thermal and af cleaning we conclude that the NRM of the Alba formation in the Villasecino area has a unique stable component with blocking temperature in the range 300°-630°C and coercivity of 100 mT. Instability during heating makes it difficult to obtain high precision on the direction; for this purpose, af cleaning is more efficient. It will be shown later that after unfolding the mean directions given by both af and thermal cleaning are almost identical but differ slightly from that of the NRM.

Chemical Demagnetization

The usefulness of chemical leaching techniques has been clearly demonstrated [Roy and Lapointe, 1978; Henry, 1979]. If in a sedimentary rock there is overlap between the blocking temperature spectra (or coercivity spectra) of two components of magnetization, these two components cannot be distinguished easily. With chemical leaching (generally HCl for red sandstone) the corresponding leaching time spectra for the two components may differ significantly enough to allow a clear separation. Failure to obtain a multivectorial origin for the NRM of the Alba formation using af and thermal cleaning does not mean that such an origin does not exist (because of the possibility of overlapping); thus chemical leaching was tried. Because of the composition of the rock (carbonate limestone), use of HCl results in destruction of the sample. We tried using a saturated solution of sodium sulfite (Na_2SO_3) which renders iron hydroxides soluble.

Ten samples, each with a 10 mm diameter hole drilled along the sample axis to increase diffusion of the solution, were leached for various periods of up to 800 hours in the saturated solution, but no significant decrease of magnetization intensity was registered. In a last attempt a sample was leached for 1 week in a solution heated at a constant temperature of 60°C. The color of the rock altered, and the solution became reddish, chemical analysis showing very clearly that iron hydroxides were abundant in the solution, but again there was no significant change in NRM direction and intensity. A section through the sample showed that leaching has penetrated about 1 mm into the rock. This represents about 20% by volume of the sample which should be enough to cause a significant decrease in intensity if iron hydroxides were the carriers of NRM. We conclude that they are not.

Magnetic Minerals

Goethite is the only magnetic mineral that we have identified using the optical microscope (resolution 2 μm) and X ray analysis. But goethite has a Curie point of only about 100°C [Stacey and Banerjee, 1974], whereas thermal demagnetization (Figure 4c) indicates that our samples have bloc-

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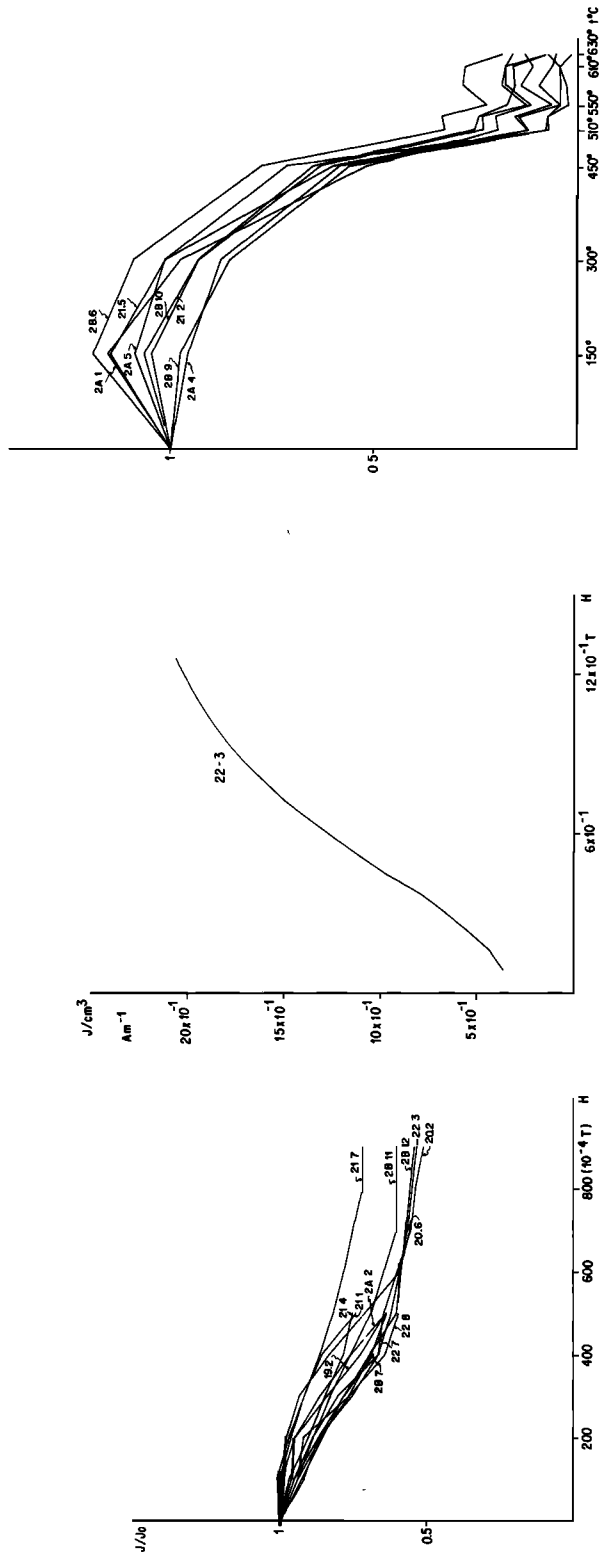


Fig. 4. (a) Alternating field demagnetization. Normalized intensity (J/J_o) versus peak field value (H) for all sites. (b) Isothermal remanent magnetization (IRM acquisition curve for sample 22-3). (c) Thermal demagnetization. Normalized intensity (J/J_o) versus temperature (t) for eight samples form sites 2A, 2B, 21.

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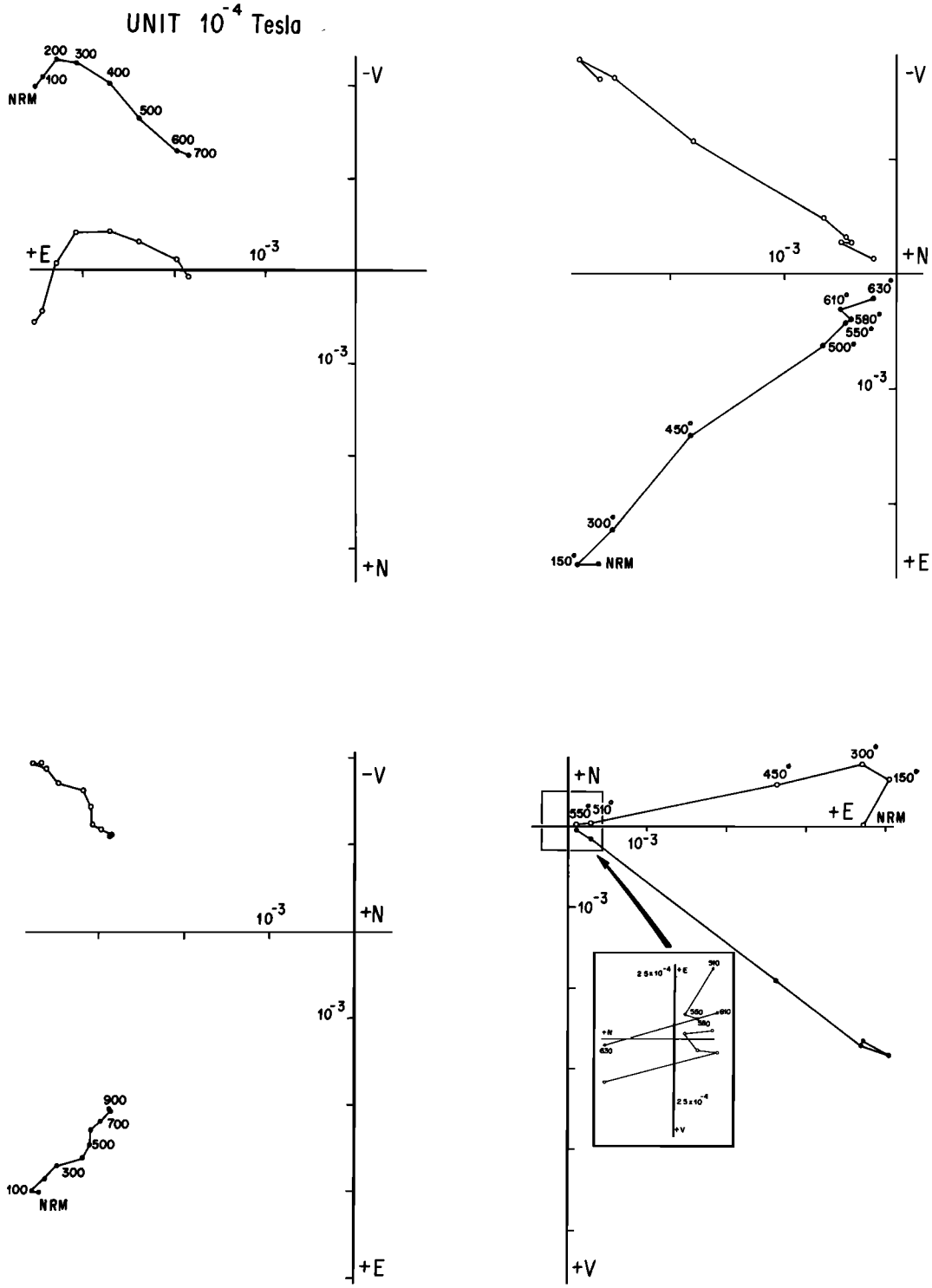


Fig. 5. Zijderveld diagrams for alternating-field demagnetization (left) and thermal demagnetization (right) for two characteristic samples (top and bottom). Open circles indicate vertical projection; full circles, horizontal projection.

king temperatures of 450°-500°C. Also chemical demagnetization suggests that goethite is not the main carrier of the NRM.

Other likely candidates are hematite, magnetite, and maghemite. Maghemite has been reported from other limestones by Channel [1978] and Heller [1978]. Also it has been shown by Yariv et al. [1979] that goethite may be transformed into maghemite by heating to 300°C in the presence of organic materials. Such high temperatures were probably not attained in the Alba griotte, but the same transformation might conceivably have occured at lower temperatures, given the geological time available.

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To investigate this, the isothermal remanent magnetism (IRM) was measured for one sample (22-3) magnetized in fields up to 1.2 T. (Figure 4b). Since no saturation was observed, we can exclude magnetite and perhaps maghemite as the main magnetic minerals but cannot otherwise discriminate between goethite and hematite.

Recently one of us (H.P.) has measured the IRM of specimens from the same Alba formation but another area of the Asturian Arc, using a cryogenic magnet with fields of up to 5 T. Of the acquired IRM, more than 80% remained after subsequent heating to 120°C. This is higher than the Curie point of goethite, which we therefore exclude as the main magnetic mineral. The IRM curves also indicate that saturation is reached at a field strength too high for maghemite but reasonable for hematite. We conclude that goethite is the most abundant magnetic mineral but that hematite is probably the main carrier of the NRM.

Models for Unfolding the Villasecino Anticline

We have shown that NRM directions are relatively stable during af and thermal cleaning and also that they become slightly better grouped after a simple 'bedding correction'. This suggests that magnetization was acquired before folding. The well-grouped values of the angle between magnetization vector and bedding (Figure 3b) support this. Failure to obtain a good value for the Fisher precision parameter k after a simple bedding correction suggests, however, that this correction (referred to here and in Table 2 as model A) is perhaps too simple.

More elaborate models can be proposed on the basis of the structural history (reviewed previously) of the Villasecino area. We have tested the following six models, each involving a sequence of rigid rotations about various axes and through various angles.

Model A. This is the simple bedding correction. There is one axis of rotation, the strike of the bedding. The angle of rotation is the dip of the bedding as measured for each sample.

Model B. The rotations described for model A are preceded by a rotation through 35° about vertical axis. This correction takes into account the curvature in the western part of the area (subarea A) and is only applied to sites 20 and 21.

Models C and D. These two models take into account the two fold phases. Unfolding starts about F2 and continues about F1 (the reverse order of folding). In both models, F2 axes are those measured on-site (Table 1) or are otherwise assumed to be vertical (sites 20 and 21). Models C and D differ in that C uses bedding orientations measured for each sample, whereas D uses site mean bedding orientations.

Models E and F. These two models assume the same order of fold phases as in C and D, but instead of using F2 fold axes as measured in the field (Table 2), they assume these axes are everywhere vertical. The difference between models E and F is the same as that between C and D. It is used to check whether local variations in bedding orientation are of tectonic origin or not.

Fortran IV programs have been written and used to apply the sequences or rotations required by

TABLE 2. Comparison of Unfolding Models

	2a			2b			19			20			21			22			All Sites			
	D	I	k	D	I	k	D	I	k	D	I	k	D	I	k	D	I	k	D	I	k	α_{95}
In situ	114.3	4.4	86	70.8	7.4	34	124.4	-2.3	34	114.3	3.5	50	136.5	-24.5	65	75.9	17.9	18	106.2	1.3	7.6	26.0
A	97.4	17.8	71	76.1	4.9	28	116.2	23.5	40	126.0	24.5	44	135.9	24.9	67	96.0	19.7	18	107.3	20.4	14	-
B	97.4	17.8	71	76.1	4.9	28	116.2	23.5	40	91.0	24.5	44	100.9	24.9	67	96.0	19.7	18	96.0	19.6	31	-
C	95.6	17.8	77	81.0	4.9	30	116.5	23.5	41	101.4	24.7	42	115.1	24.9	45	95.3	19.8	17	100.5	19.7	30	12.4
D	97.0	17.9	86	82.0	8.7	34	117.1	18.1	34	99.5	27.1	50	110.9	20.5	65	98.1	18.0	18	100.6	18.7	39	11.0
E	90.2	17.7	86	96.2	5.1	23	117.2	23.5	39	101.4	24.7	42	115.1	24.9	45	97.4	19.8	17	102.7	19.6	41.0	10.6
F	91.8	17.8	86	93.0	8.6	34	112.5	17.9	34	99.5	27.1	50	110.9	20.5	65	100.6	18.2	18	101.3	18.5	64	8.5

Populations of all directions: N = 42, D = 101.5, I = + 18.5, α_{95} = 1.1, k_w = 36, k_b = 90.
Fisher statistics of NRM directions for each site and all sites before unfolding (in situ) and after unfolding using models A to F.
See Table 1 for the number of samples used at each site. All angular data are in degrees.

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the above models. The input data include the magnetization directions for each sample and the structural data measured in the field (Table 2). The output includes the magnetic vectors after unfolding and the Fisher statistics for each and all sites. The main algorithm contains the following steps:

1. Rotate bedding about local F2 axis until bedding pole is normal to F1 (that is, the bedding surface contains F1). This is done iteratively, with checking of the dot product between the two orientations. Apply the resulting rotation to magnetic vectors (models C and D). Alternatively rotate about a vertical axis (models B, E, F).

2. Restore F1 to horizontal by rotation about horizontal axis normal to F1. Apply to magnetic vectors (models C to F).

3. Rotate about F1 until bedding is horizontal (models C to F).

Notice that steps 2 and 3 above are commutative, but other pairs of steps are not necessarily so.

Models A to F were first tested using a maximum number of samples, that is, all 42 original NRM directions. This was judged preferable to using only the 13 magnetically cleaned samples or only the 17 thermally cleaned ones. In any case, the within-site scatter of directions before or after cleaning is small compared with the between-site scatter due to folding.

Fisher statistics for each and all sites (Table 2) were calculated after applying each of model A to F. Using first the results for each site, we can examine the effect of local variations of bedding orientation. These variations are not negligible: for example, for site 19, the range of strike directions is 16°. Bearing in mind that the site diameter is only about 10 m, we can ask whether significantly different results are obtained using either (1) the bedding orientations measured for each sample, or (2) the site mean bedding orientation. Each column of Table 2 shows that (1) k values do not vary much from one model to the next, (2) except for site 19, k nevertheless increases from model C to model D and from E to F: in other words, k is systematically slightly higher when site mean bedding orientations are used. One explanation for this is that bedding variations are, in general, due not to deformation but to some previous process such as nonplanar deposition. For site 19, which is near an F2 fold hinge, all models give a slight increase in k when using sample bedding. This might be due to deformation. On a strict Fisher statistical basis, however, the k values at each site are not significantly different from one model to the next. This is confirmed by a Watson test.

If we consider now the result for all sites (last column, Table 2), the various models can be compared. The striking feature of the results is that progressively higher values of k are obtained for progressively more elaborate models. Thus model A is by far the least satisfactory (k = 14) model for unfolding the structure. This mean that the classic bedding correction must be applied with caution and only in relatively simple tectonic settings, certainly not in complexly folded areas like Cantabria. Better results are obtained with model B (k = 31) which takes into account bending about a vertical axis. This tends to confirm that the NRM is pretectonic and suggests that the bending is a late feature. Far better are the models which consider two folding episodes, F1 and F2. In particular, the best is model F (k = 64), which assumes that F2 is vertical.

Structural mapping and analysis independently suggest that models C to F are the most reasonable (see previous discussion). Models C and D imply both upward and lateral bending of F1 axes, whereas models E and F imply only lateral bending. Lateral bending about vertical axes is certainly prevalent in Cantabria [Julivert and Marcos, 1973], but upward bending also exists. The field evidence therefore favors models C and D, or something between these and the more extreme E and F. Although the unfolding ot the paleomagnetism favors model F, the evidence is probably not sensitive enough to enable a firm choice to be made.

Direction of Earth's Field at Villasecino During the Lower Carboniferous

So far we have been using undemagnetized NRM data. The main goal, a comparison of different unfolding models, justifies this method. We can now use model F to correct magnetic directions for folding and check the efficiency of af and thermal cleaning. Model F eliminates most of between-site variation due to cross folding of the structure. Results are given in Table 3 for each site and all sites, for af cleaned samples and thermally cleaned samples. We draw attention to the following points:

1. For each site the Fisher parameter k is higher after cleaning than it is before; for example, for site 2B, k = 34 before (Table 2) and k = 116 after (Table 3). This shows the efficiency of cleaning; removal of secondary components results in improved clustering of directions for each of the six sites.

2. For the all-sites results (Table 3, last column) the site mean direction is almost the same after either af cleaning (D = 100.8, I = 14.4) or thermal cleaning (D = 102.9, I = 12.7). This confirms that there is a unique stable component direction in the NRM, as suggested by demagnetizing curves.

3. The inclination (14.1) obtained after cleaning for all six sites is, however, shallower than that (18.5°, Table 2, model F) obtained before cleaning.

4. The af cleaning (k = 81 for all sites) gives better precision than thermal cleaning (k = 32) on the mean direction.

Our results show that af cleaning not only reduces all-site scatter (k = 64 for NRM, Table 2, and k = 81, Table 3) but contributes greatly to reducing between-site scatter. On the basis we could use the 13 af cleaned directions to give the best estimate direction of magnetization of the rock before cross-folding. However, because the site mean direction is almost the same after af or thermal cleaning, there is no reason to ignore the second set of data, and we will estimate the mean direction of the earth's field in the lower Carboniferous at Villasecino using all 30 cleaned directions.

To obtain the overall mean direction after

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TABLE 3. Demagnetized and Unfolded Data

Sites	af Demagnetization				Thermal Demagnetization				all Demagnetization			
	D	I	k	α_{95}	D	I	k	α_{95}	D	I	k	α_{95}
2a	93.1	14.5			101.9	17.1			99.7	16.5	71.6	8.3
2b	94.0	4.3			95.6	-3.3			94.8	0.5	116.4	5.3
19	106.6	13.8			120.0	23.4			116.5	21.1	92.4	7.3
20	100.6	17.4			96.2	15.0			97.9	16.0	246	4.0
21	110.1	23.5			110.2	20.3			110.1	22.2	202	4.4
22	100.8	12.1			95.3	2.5			98.0	7.3	50.6	8.0
All	100.8	14.4	81	6.5	102.9	12.7	32	11	102.6	14.1	48.5	8.5

Fisher statistics for each site and all sites are given after unfolding using model F. All angular data are in degrees.

Population of all demagnetized samples: N = 30, D = 101.8, I = +13.1, α_{95} = 1.5, k_w = 94, k_b = 53 (all cleaned directions but taking account of between-site and within-site scatter in a two-tier analysis).

unfolding, we have been using site mean directions so far (see Tables 1 and 3, 'all sites'). The choice is sound if we look at site size (∿10 m) compared with anticline size (∿3 km). Alternatively, the magnetic direction of the formation can be estimated using the vector résultant of all samples (bottom rows of Table 2 and 3). The two mean directions obtained agree closely, and the difference results from problems of estimating the error in the mean direction. A two-tier analysis [see McElhinny, 1973] will take into account within-site variation and between-site variation; if we apply this to both populations (42 NRM directions, and 30 cleaned characteristic directions), we obtain the following results.

1. The mean direction for NRM is D = 101.5°, I = 18.5° with within-site Fisher precision parameter k_w = 36 and between-site k_B = 90. This leads to a mean direction given with precision α_{95} = 1.1°.

2. For the 30 cleaned directions the mean is D = 101.8°, I = 13.1°; within-site k_w = 94 and between-site k_B = 53. The mean direction is given with α_{95} = 1.5°.

3. By comparing the Fisher parameters for both populations we may conclude that the two mean directions obtained are significantly different.

We conclude that the mean direction D = 101.8° I = +13.1°, α_{95} = 1.5° obtained using the 30 cleaned directions of magnetization and unfolding with model F is the best estimate of the direction of the earth's magnetic field at Villasecino during the lower Carboniferous.

Tectonic Implications

It is useful to compare the result obtained in the last section with Carboniferous paleomagnetic data available for 'stable' Europe [Morel and Irving, 1978]. To do this we have to allow for the late Cretaceous rotation of Spain (through an angle which we will denote as RS). This rotation was first demonstrated by paleomagnetism [Van der Voo, 1967, 1969; Van den Berg, 1979] and later confirmed using marine magnetic anomalies [Le Borgne et al., 1971] and structural features [Le Pichon and Sibuet, 1971; Le Pichon et al., 1977]. If we choose Le Pichon's 'Paris' pole of rotation (50°N, 3.3°E), an anticlockwise rotation of RS = 23°, and San Emiliano as the reference site for the Villasecino area, we obtain new coordinates (45.8°N, 9.5°W) for San Emiliano before the Cretaceous and a new paleomagnetic direction for Villasecino, D = 122.0°, I = 13.1°. Alternatively, using the model of Carey [1958], we obtain D = 137°E, I = 13.1°. If we now take a value (38°N, 161°E) for the Carboniferous paléomagnetic pole for Europe [McElhinny, 1973], we can calculate the direction of the earth's field at San Emiliano at that time: it is D = 187°5 E, I = 11.4° or, alternatively, D = 7.5° E, I = -11.4° (magnetic reversal). The calculate inclination (11.4°) is very close to the value (13.1°) obtained from paleomagnetic samples after cleaning and unfolding.

In contrast, the calculated declination (187.5°) differs by 65° (Le Pichon's pole) or 50° (Carey's pole) from the value obtained after cleaning and unfolding. This suggests that the Villasecino area has undergone an additional rigid rotation, RV, clockwise or anticlockwise, about a vertical axis. If RV has been clockwise, the area must originally have been in northerly latitudes; if anticlokwise, in southerly latitudes (7° S). We prefer the second alternative for several reasons. First, red beds of Normandy and Brittany with paleomagnetism dated as Late Devonian to early Carboniferous give a paleolatitude of 8° S for this Armorican region [Jones et al., 1979]. Second, if RV is a local rotation affecting the Villasecino area but not necessarily the whole of Cantabria, the sense must be compatible with the resulting curvature of the Asturian arc: RV must have been anticlockwise. Third, this agrees with paleomagnetic results obtained recently by Ries et al. [1980].

Notice that an anticlockwise rotation RV of the southern part of the Asturian Arc, relative to the northern part, is compatible with all the models so far proposed for the secondary origin of the arc. Thus our paleomagnetic results do not discriminate between the model of an orocline by bending [Ries and Shackleton, 1976] and the model of indentation by a westerly moving microplate [Matte and Ribeiro, 1975]. Paleomagnetism does, however, suggest that the arc is partly secondary in origin.

Conclusions

From our paleomagnetic and structural work we draw the following conclusions concerning the history of the Villasecino area in particular and of Cantabria in general.

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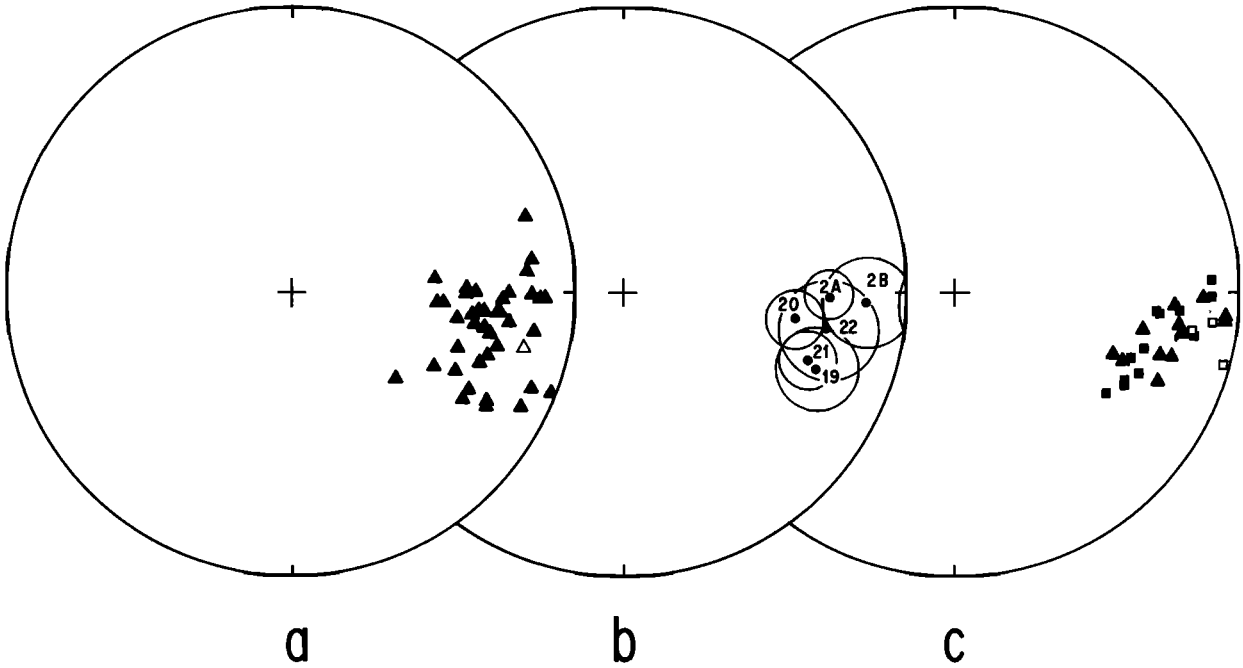


Fig. 6. Stereographic projection of unfolded magnetization (model F). (a) NRM directions (solid triangles = lower hemisphere; open triangle = upper hemisphere); (b) NRM directions (site means with circles of confidence). (c) Demagnetized directions (squares = thermal; triangles = alternating field; open symbols = upper hemisphere; solid symbols = lower hemisphere).

1. Magnetization components now observed in the Alba formation were formed after sedimentation (Viséan) and mainly before folding (Westphalian). The carrier mineral is probably fine-grained hematite.

2. Folding occured in two successive phases. Locally, the first phase was about a nearly horizontal east-west axis, the second about a nearly vertical axis. Folding of the Alba formation involved rigid rotations with little or no internal strain. Regionally, the second folds have axial surfaces radial to the Asturian Arc: they may have formed during tectonic closure of the arc.

3. The Villasecino area underwent an anticlockwise rotation (50° to 65°) about a vertical axis. This is tentatively correlated with formation or accentuation of the Asturian Arc by tectonic processes.

We believe our results emphasize the need for paleomagnetic sampling to be combined with careful structural mapping, especially in moderately complex orogenic areas such as Cantabria. Even more care would certainly be required in areas where the rocks have undergone intense penetrative strain as well as rigid rotations.

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